AI-Supported Numeracy Teaching

Entry tasks: Access and complete each of the below while we are waiting

Share your readiness

Complete the Microsoft Form below to share your responses from the online Readiness section (you may update your answers if they have changed). Don't overthink your responses.

https://forms.cloud.microsoft/r/rNEZfgGzQD

Share your perspective

Open the Padlet below and write a post that includes the following:

- Your personal definition of numeracy
- Why you feel it is important that numeracy is taught as part of your specialisation
  (subject area / phase of learning)

https://tinyurl.com/5367hasj

Enhancing Literacy and Numeracy Skills Using AI · Module 3

AI-Supported Numeracy Teaching

Using generative AI to generate, evaluate and refine numeracy tasks

Blake Segler · School of Education, Curtin University



Ground

Today: what we will do

- Ground: your definition of numeracy, and numeracy in different countries
- Readiness: what you told us in the online part
- Concept 1: how generative AI handles numeracy questions
- Concept 2: using AI to generate and evaluate task ideas
- Concept 3: using AI to produce resources
- Express: what worked, what did not

The main output: a fully resourced numeracy task for a particular context

Ground

Your “Readiness” ratings:

Rate each element: your confidence, your confidence in AI, and your efficiency.

- Personal numeracy capability
- Identifying numeracy opportunities in your phase or learning area
- Developing resources
- Providing differentiation and scaffolding
- Assessing numeracy
- Explaining ideas and choosing representations
- Giving feedback to learners



https://forms.cloud.microsoft/r/rNEZfgGzQD

Ground

Numeracy in three countries

Australia

A capability spread across the curriculum, and part of the Mathematics learning area.

Japan

Part of mathematics itself, taught through structured problem-solving lessons.

South Africa

A subject in its own right: Mathematical Literacy.

For each: how is numeracy interpreted, who is responsible for teaching it, and what approaches dominate?

Goos & O'Sullivan (2023)

Ground

How does Indonesia compare?

If an AI drew mainly on one of these countries, how appropriate would it be for your teachers and students?

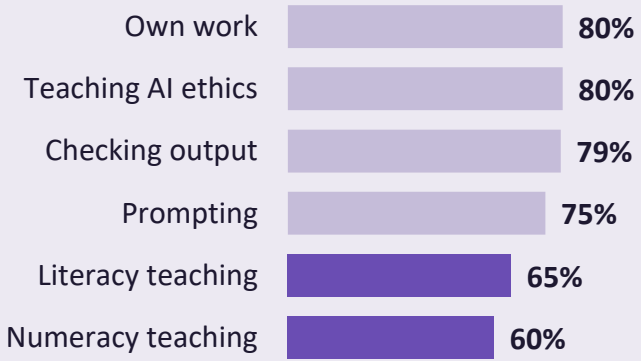
- How do these three positions contrast with Indonesia?
- Use the Padlet posts: where do your definitions fit?
- What could go wrong if the AI’s training data does not match your context?

Readiness

What we heard from you

Confidence using AI

% fairly or very confident, by area



n = 40; 14 teach numeracy or maths

Key considerations

AI's fit with Indonesian context is the lowest-rated item (**3.5/5**); 18 of 40 were neutral.

Opportunities

31 of 40 agree AI can improve numeracy learning. Already used for materials (32) and quizzes (31).

Concerns

Over-reliance weakening critical thinking (**35 of 40**) and inaccurate or made-up output (**32 of 40**).

Pre-programme survey. How do your "Readiness" ratings for this session compare?

Concept 1: Generative AI and numeracy

What happens when we give AI numeracy questions?

Autonomy

Prompt 1

“What is 48271 x 9386?”

- Was the AI response correct?
- How does AI reach an answer, compared with how you would?
- Why do we still need a human to evaluate it?

Prompt 2

“What is 2.300,456+500,412?”

- Did the AI read the numbers the way Indonesian notation writes them?
- How might its training data lead it to a wrong, or inappropriate, answer in a local context?

Autonomy

Prompt 3

“Two shops offer rice at 3 kg for Rp4.000.000 and 5 kg for Rp9.000.000. Which is the better buy? Explain in a way suitable for a Kelas 6 learner.”

- Did the AI notice that the prices are unrealistic?
- Is this the reasoning you would encourage students to use?
- Is this the type of explanation you would use with students?

Autonomy

What might strong numeracy look like here?

A 5 kg bag costs Rp9.000.000. Two 3 kg bags cost Rp8.000.000 and give 6 kg.

Less cost and more rice. No unit prices needed.

Did any AI response reason like this?

Autonomy

Prompt 4

“Suggest a meaningful numeracy problem for a Kelas 3 child in Indonesia.”

- Is it appropriate and meaningful for most students in your context?
- In what ways might it not be?
- Talk, or add your view in the chat.

Competency

The human in the driver's seat

Guide the AI

Give appropriate guidance: context, purpose, constraints.

Evaluate the output

Check accuracy, pedagogy and context. Do not accept it as given.

You stay in charge of the thinking

Roles

Evaluating my relationship with AI

In engaging critically with AI, I find it useful to consider how I am positioning the AI – what professional role is it most akin to?

I then apply similar professional rules for how I would respond



Supervisor

Not how I use AI



Colleague

Perspectives, feedback, ideas



Assistant

Admin and routine tasks

Roles

Evaluating my relationship with AI

In engaging critically with AI, I find it useful to consider how I am positioning the AI – what professional role is it most akin to?

I then apply similar professional rules for how I would respond



Supervisor

Not how I use AI



Colleague

Perspectives, feedback, ideas



Assistant

Admin and routine tasks



Contract Worker

Tasks where I have limited skill

Roles

Evaluating my relationship with AI

In engaging critically with AI, I find it useful to consider how I am positioning the AI – what professional role is it most akin to?

I then apply similar professional rules for how I would respond



Supervisor

Not how I use AI



Colleague

Perspectives, feedback, ideas

What do I want to take on board / pursue further?



Assistant

Admin and routine tasks



Contract Worker

Tasks where I have limited skill

What is fit for purpose?

What do I want re-attempted with clarification?

What do I want to re-assign?

Workflow

A practical framework

1. Direct
Set audience, purpose, tone, constraints

→

2. Generate
AI produces a first draft or set of options

→

3. Evaluate
Check for accuracy, bias, fitness for purpose

→

4. Refine
Iterate — trim, reframe, regenerate specific parts

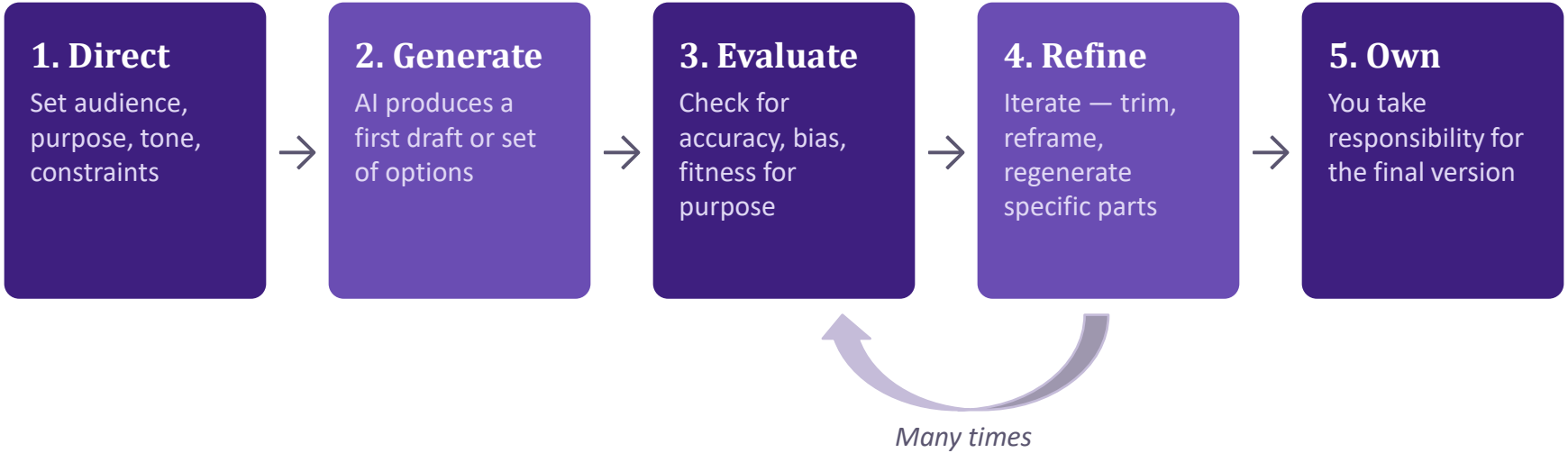
→

5. Own
You take responsibility for the final version

You direct AI assists

Workflow

More practical frameworks

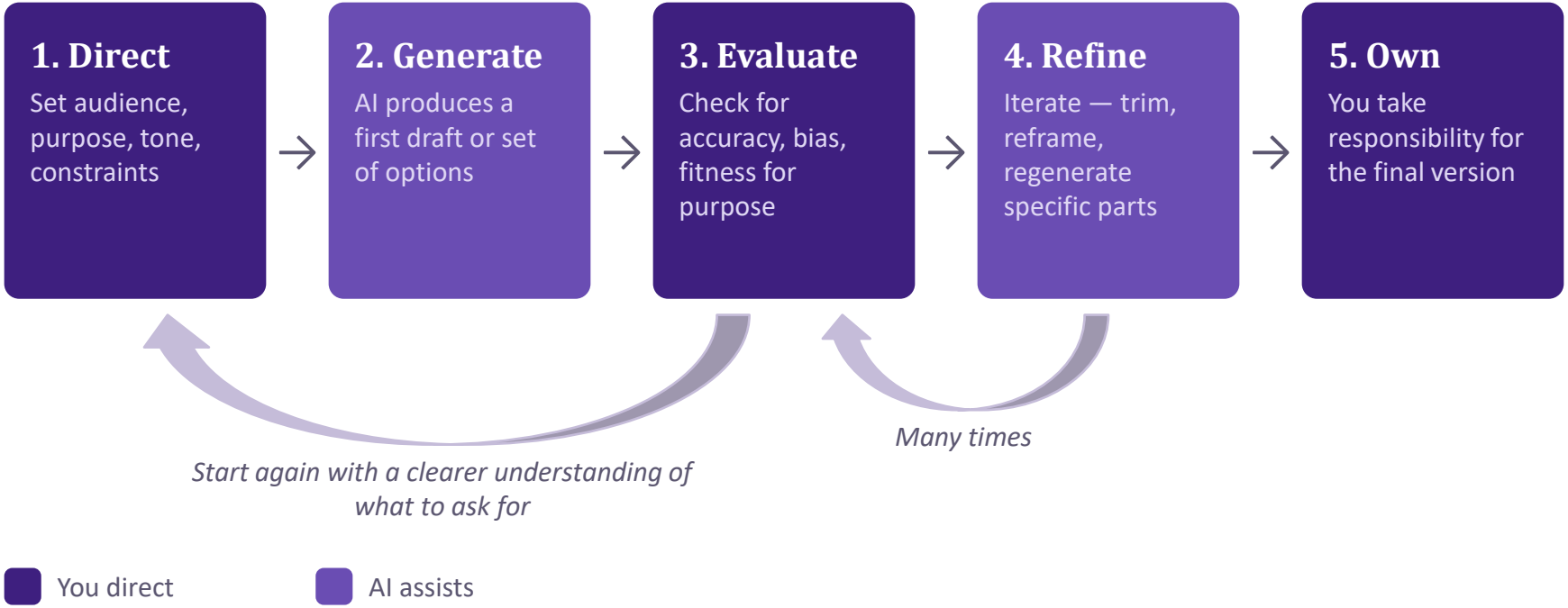


You direct

AI assists

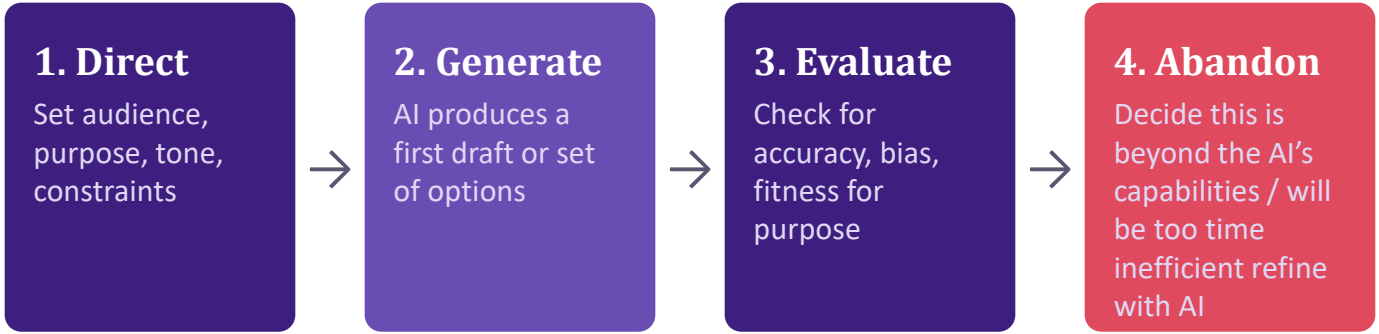
Workflow

More practical frameworks



Workflow

More practical frameworks

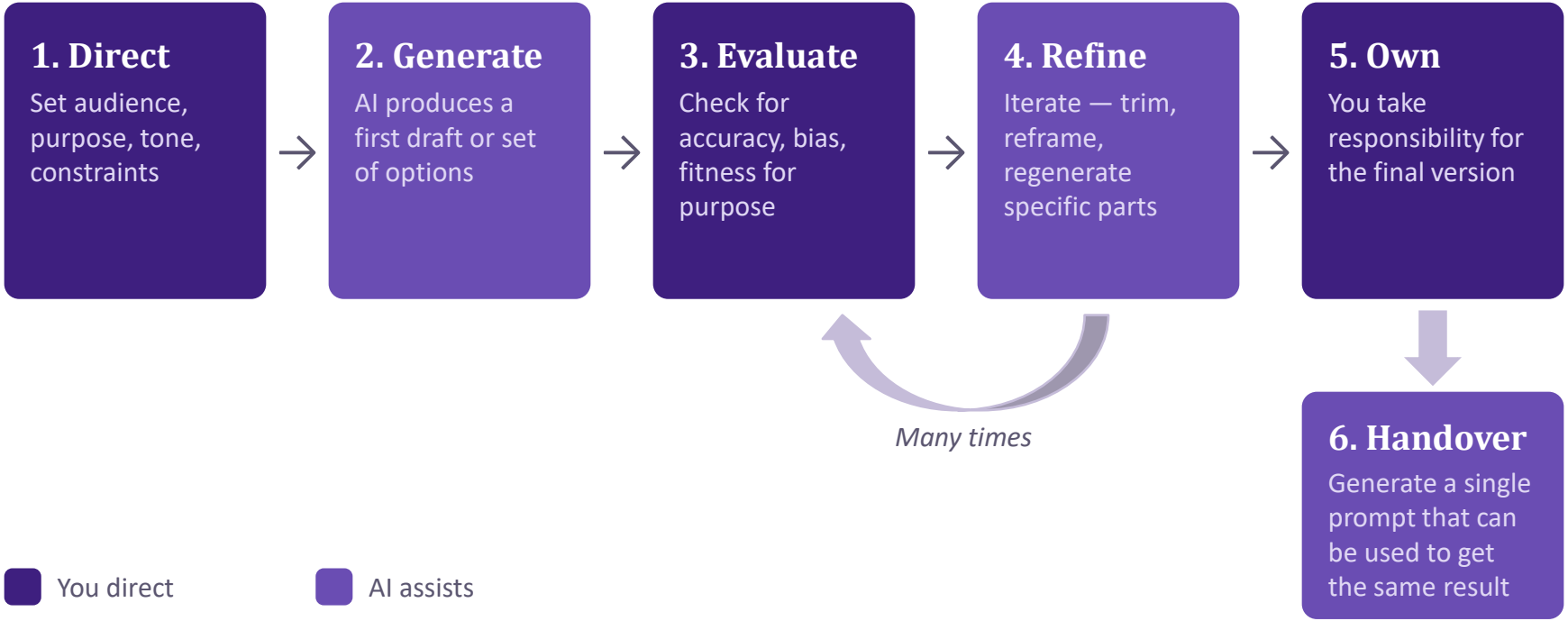


You direct

AI assists

Workflow

More practical frameworks



Workflow

Prompts

1. Direct

Set audience, purpose, tone, constraints

You direct

The degree of prompt engineering I employ is a balancing of effort against complexity and importance

For more complex/important tasks I put more effort into things like:

- Giving more context (audience, purpose, tone)
- Giving intentions (requirements, examples/templates, preferences)

as well as:

- Assigning roles / perspectives
- Explicitly asking what it would like clarified
- Inviting alternatives
- Meta prompts

Concept 2: Using generative AI for ideation

Working towards a numeracy task for your context

Breakout 1: share your prompt and your evaluation

In groups of two or three, share how you designed your prompt in the online component of this module and how you evaluated the output. Think about:

- Parts of the output that did not match your prompt
- Parts that did not meet your intention, even if you did not put them in the prompt
- Anything inappropriate or sub-optimal

Competency

Improving your prompt

- Relevant curriculum documents
- Expected task duration
- Scope of the task
- Clearer assessment criteria
- Expectations and limitations of students
- Focus of the task

Ask for a list of possibilities, for example 10 tasks. Then review and choose.
You may also wish to compare the outputs of different AI agents.

Competency

The three elements of numeracy in Indonesia

Konten

Students' knowledge of relevant content.

Proses kognitif

Students' cognitive processes, including reasoning and justification.

Konteks

Application to meaningful, real-life contexts.

Also: does the task align with the curriculum of your learning area?

Asesmen Kompetensi Minimum (AKM), Asesmen Nasional

Competency

Evaluating a list of tasks

Appropriate

Ready to use as it is.

Appropriate after modifications

A good idea that needs changes. Say what changes.

Not appropriate

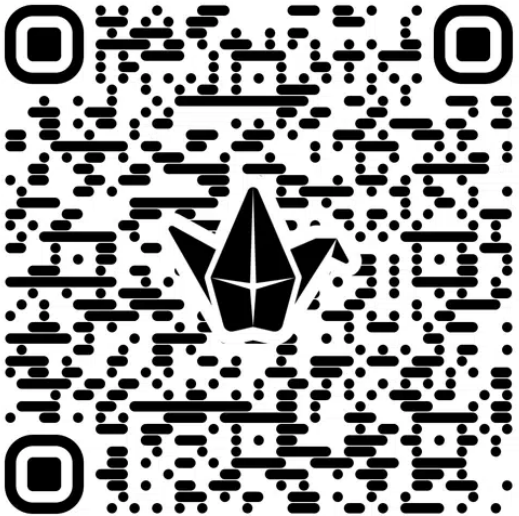
Does not suit your students or purpose.

Use your criteria: assumed knowledge, relevance, and opportunities to apply reasoning.

Competency

Breakout 2: refine, generate, review, and choose

- Refine your prompt
- Generate a list of 5 possible numeracy problems
- Review each: appropriate, appropriate after modifications, or not appropriate
- Choose one. Justify it, or suggest modifications (with reasons)
- Add to the Padlet table: your prompt, chosen task, strengths, ideas for improvement



https://tinyurl.com/2vcw3hv5

- Use your breakout room partner(s) to discuss your choices and receive feedback
- Let me know if you need assistance!

Competency

Back in the main room: compare

The original single suggestion

One task from one prompt.

A task chosen from a list of 10

A wider range, reviewed against your criteria.

Your professional expertise still finds the improvements and guides the AI

Concept 3: Using generative AI to produce resources

From a good task to a usable resource

Resourcing and differentiation

- Look back at your online ratings for developing resources and for differentiation and scaffolding
- Where is your own efficiency limited?
- Where could AI help, and what are its limits?

Competency

What AI can generate quickly

Student templates

Also in differentiated versions.

Rubrics

Self-assessment and teacher-assessment, aligned with your criteria.

Example student responses

To see what answers may look like.

Optional: support materials for students who lack the content knowledge the problem needs, and teacher guidance notes.

Competency

Breakout 3: generate, evaluate, and refine

- Prompt AI for the resources you need
- Evaluate each output against your pre-work considerations
- Refine by further prompting, or by editing it yourself
- Add to the Padlet table: the current state of each resource, and improvements still to make



https://tinyurl.com/3x67y5ev

Express

Where was AI effective, and where was it a challenge?

Add to the Padlet: one column for where AI was effective, one for challenges

Think about what you produced and how you got there.

https://tinyurl.com/9cfvrd7m

Express

Prompt again or take control?

- When was continued prompting the best way forward?
- When would it have been better to take control yourself?

Express

Further reflections

- What is the teacher’s role in using these resources well with learners?
- What improvements should still be made?
- How do the range of ideas and the efficiency compare with and without AI?

Express

Carry forward

Your numeracy problem, prompts, and reflections feed into your lesson plans

The recording and these slides will be added to the online page after the session.